



Statistical Data Science

CSE 303

Project Title

Credit Card Fraud Detection

Submitted To

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Section: 8

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Credit Card Fraud Detection

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1. Dataset Description

1.1 Source and Real-World Problem

The dataset used in this project is the Credit Card Fraud Detection dataset published by the Machine Learning Group at Université Libre de Bruxelles (MLG-ULB) and available on Kaggle at: [Credit Card Fraud Detection](#).

It addresses one of the most pressing challenges in the financial industry — the automated detection of fraudulent credit card transactions in real time. The data covers two days of transactions made by European cardholders in September 2013. Due to confidentiality, the original features have been transformed using PCA, producing 28 anonymized components (V1–V28). Only Time and Amount retain their original meaning.

1.2 Dataset Overview

Property	Detail
Total Observations	284,807 rows
Total Features	31 columns (30 input features + 1 target variable)
Source	Kaggle — MLG-ULB (Université Libre de Bruxelles)
Time Period	September 2013 — Two days of European cardholder transactions
Real-World Problem	Detection of fraudulent credit card transactions
Fraud Cases	492 (0.1727% of all transactions)

Table 1.1: Dataset Overview Summary

1.3 Feature Description

The table below describes each feature, its type, meaning, and classification.

Feature	Type	Description	Classification
Time	Numerical (Continuous)	Seconds elapsed since the first transaction in the dataset	Numerical – Continuous
V1–V28	Numerical (Continuous)	Anonymized PCA-transformed components (original features confidential)	Numerical – Continuous
Amount	Numerical (Continuous)	Transaction value in Euro (EUR)	Numerical – Continuous
Class	Categorical (Binary)	0 = Legitimate transaction, 1 = Fraudulent transaction	Categorical – Binary

Table 1.2: Feature Description of the Credit Card Fraud Detection Dataset

2. Central Tendency and Dispersion Analysis

The table below presents a full statistical summary for all 31 features: mean, median, mode, variance, standard deviation, Q1, Q3, IQR, skewness direction, and outlier count using the IQR rule.

- All V1–V28 PCA features have a mean ≈ 0 , as expected from PCA standardisation.
- Amount is highly right-skewed (skewness = 16.98); mean EUR 88.35 far exceeds median EUR 22.00, indicating most transactions are small while a few are very large.
- Time shows near-zero skewness (-0.04) but a bimodal distribution reflecting day/night transaction cycles.
- Class has near-zero variance (0.00) as 99.83% of values equal 0.

Feature	Mean	Median	Mode	Variance	Std Dev	Q1	Q3	IQR	Skewness	Outliers
Time	94813.87	84692.00	163152.00	2.26e+09	47488.16	54201.50	139320.50	85119.00	Right-skewed	0
V1	0.00	0.02	1.25	3.84	1.96	-0.92	1.32	2.24	Left-skewed	7,062
V2	0.00	0.07	-0.33	2.73	1.65	-0.60	0.80	1.40	Left-skewed	13,526
V3	0.00	0.18	-2.75	2.30	1.52	-0.89	1.03	1.92	Left-skewed	3,363
V4	0.00	-0.02	-0.84	2.00	1.42	-0.85	0.74	1.59	Right-skewed	11,148
V5	0.00	-0.05	-0.56	1.91	1.38	-0.69	0.61	1.30	Right-skewed	12,295
V6	0.00	-0.27	-1.01	1.77	1.33	-0.77	0.40	1.17	Right-skewed	22,965
V7	0.00	0.04	-0.43	1.53	1.24	-0.55	0.57	1.12	Left-skewed	8,948
V8	0.00	0.02	-0.16	1.43	1.19	-0.21	0.33	0.54	Left-skewed	24,134
V9	0.00	-0.05	0.17	1.21	1.10	-0.64	0.60	1.24	Right-skewed	8,283
V10	0.00	-0.09	-0.08	1.19	1.09	-0.54	0.45	0.99	Right-skewed	9,496
V11	0.00	-0.03	-0.36	1.04	1.02	-0.76	0.74	1.50	Right-skewed	780
V12	0.00	0.14	-0.07	1.00	1.00	-0.41	0.62	1.02	Left-skewed	15,348
V13	0.00	-0.01	-0.52	0.99	1.00	-0.65	0.66	1.31	Right-skewed	3,368
V14	0.00	0.05	0.41	0.92	0.96	-0.43	0.49	0.92	Left-skewed	14,149
V15	0.00	0.05	1.12	0.84	0.92	-0.58	0.65	1.23	Left-skewed	2,894
V16	0.00	0.07	-0.37	0.77	0.88	-0.47	0.52	0.99	Left-skewed	8,184
V17	0.00	-0.07	-0.60	0.72	0.85	-0.48	0.40	0.88	Right-skewed	7,420
V18	0.00	0.00	-0.44	0.70	0.84	-0.50	0.50	1.00	Right-skewed	7,533
V19	0.00	0.00	-0.33	0.66	0.81	-0.46	0.46	0.92	Left-skewed	10,205
V20	0.00	-0.06	-0.18	0.59	0.77	-0.21	0.13	0.34	Right-skewed	27,770

V21	0.00	-0.03	-0.26	0.54	0.73	-0.23	0.19	0.41	Right-skewed	14,497
V22	0.00	0.01	-0.82	0.53	0.73	-0.54	0.53	1.07	Left-skewed	1,317
V23	0.00	-0.01	0.02	0.39	0.62	-0.16	0.15	0.31	Right-skewed	18,541
V24	0.00	0.04	0.36	0.37	0.61	-0.35	0.44	0.79	Left-skewed	4,774
V25	0.00	0.02	0.19	0.27	0.52	-0.32	0.35	0.67	Left-skewed	5,367
V26	0.00	-0.05	-0.40	0.23	0.48	-0.33	0.24	0.57	Right-skewed	5,596
V27	0.00	0.00	-0.04	0.16	0.40	-0.07	0.09	0.16	Left-skewed	39,163
V28	0.00	0.01	-0.06	0.11	0.33	-0.05	0.08	0.13	Left-skewed	30,342
Amount	88.35	22.00	1.00	62,560.07	250.12	5.60	77.16	71.56	Right-skewed	31,904
Class	0.00	0.00	0.00	0.00	0.04	0.00	0.00	0.00	Right-skewed	492

Table 2.1: Central Tendency and Dispersion Summary for All Numerical Features

3. Distribution and Shape

3.1 Distribution of Transaction Amount

The histogram, density plot, and boxplot below reveal the distribution characteristics of the Amount feature.

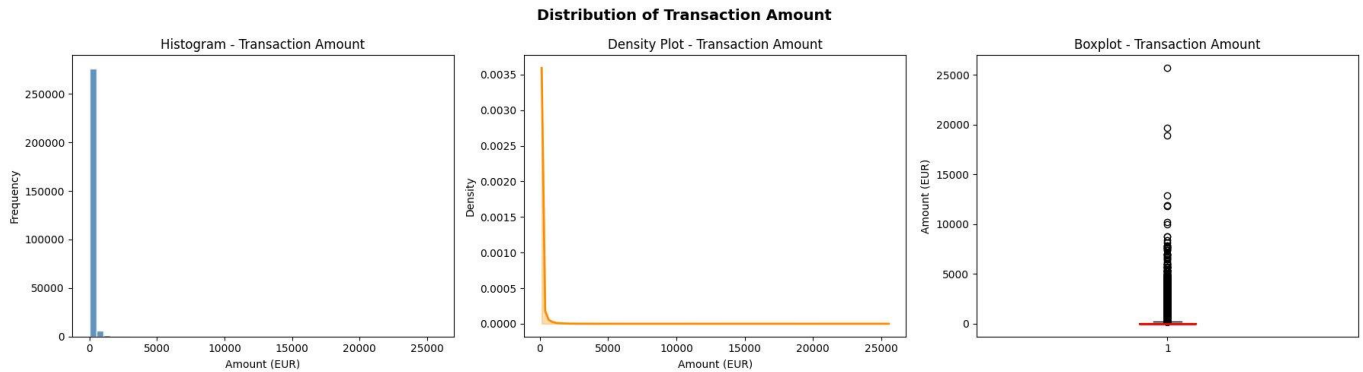


Figure 3.1: Histogram, Density Plot, and Boxplot of Transaction Amount

Interpretation

The histogram reveals a sharp spike near zero with a very long right tail, confirming an extreme right-skewed distribution (skewness = 16.98). The median (EUR 22.00) is far below the mean (EUR 88.35). The density plot shows exponential decay from a peak near EUR 0, and the boxplot highlights numerous high-value outliers reaching EUR 25,691 — all above the IQR upper bound of EUR 184.51. A total of 31,904 outliers (11.20%) were flagged. Log transformation is strongly recommended before modeling.

3.2 Distribution of Transaction Time

The histogram, density plot, and boxplot below reveal the distribution characteristics of the Time feature.

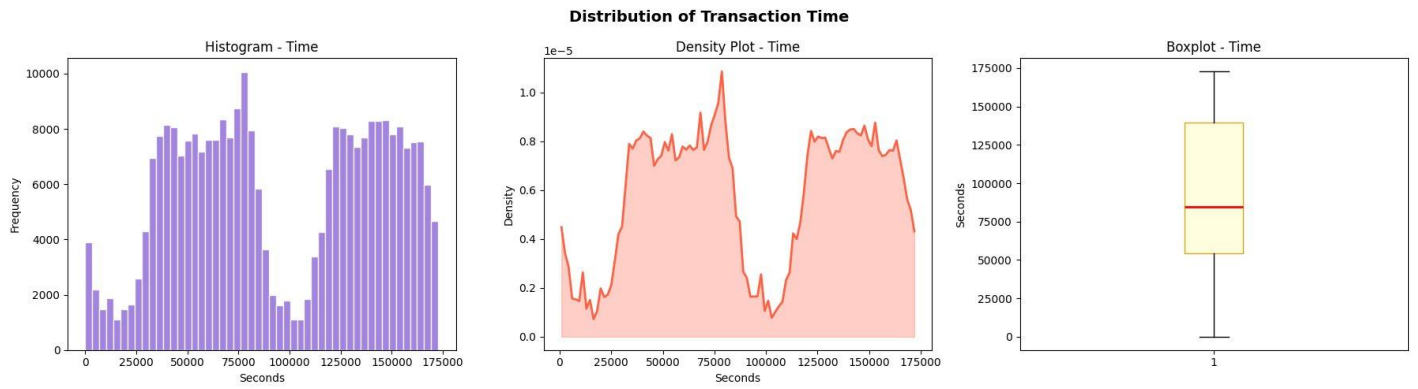


Figure 3.2: Histogram, Density Plot, and Boxplot of Transaction Time

Interpretation

Time exhibits a clear bimodal distribution with two visible peaks corresponding to daytime and overnight transaction cycles across the 48-hour period. The skewness of -0.04 confirms near-symmetry overall. The boxplot shows no extreme outliers (IQR rule yields 0 outliers), with values spread smoothly between 0 and $\sim 172,792$ seconds, confirming that transaction timing follows a natural human behavioral pattern.

3.3 Distribution of Selected PCA Features

Six representative PCA features (V1, V2, V3, V4, V10, V14) are visualized below using histograms overlaid with density curves.

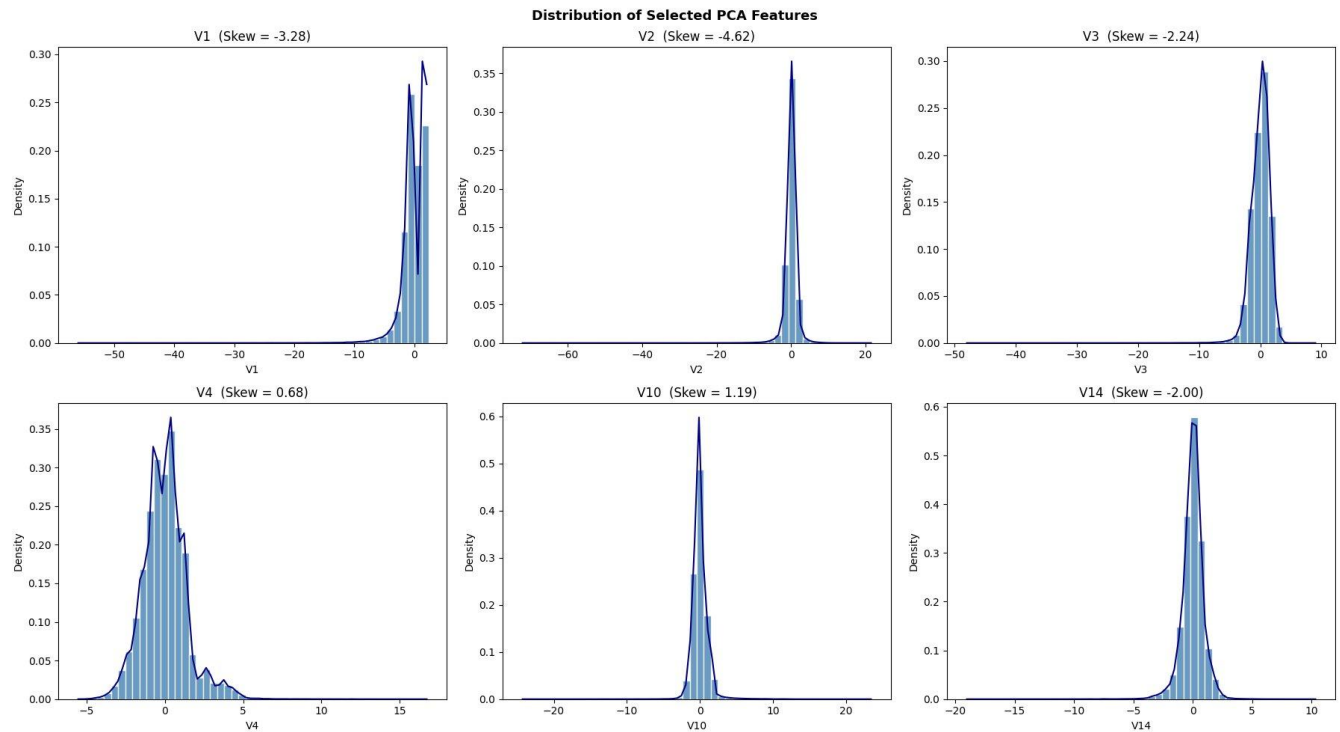


Figure 3.3: Distribution of Selected PCA Features (V1, V2, V3, V4, V10, V14)

Interpretation

Most PCA features are approximately bell-shaped and centered near 0, as expected from PCA standardisation. V1 (skew = -3.28), V2 (skew = -4.62), V3 (skew = -2.24), and V14 (skew = -2.00) are notably left-skewed with long tails toward extreme negative values driven predominantly by fraudulent transactions. V4 (skew = $+0.68$) and V10 (skew = $+1.19$) show mild right-skewness. The heavy-tailed features carry the most discriminative information for fraud detection.

3.4 Outlier Detection Using the IQR Rule

Outliers are identified using the IQR rule: values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ are flagged as outliers. The table is sorted by outlier percentage in descending order.

Feature	Lower Bound	Upper Bound	Num Outliers	Outlier %
V27	-0.31	0.33	39,163	13.75%
Amount	-101.75	184.51	31,904	11.20%
V28	-0.25	0.28	30,342	10.65%
V20	-0.73	0.65	27,770	9.75%
V8	-1.01	1.13	24,134	8.47%
V6	-2.52	2.15	22,965	8.06%
V23	-0.63	0.61	18,541	6.51%
V12	-1.94	2.15	15,348	5.39%
V21	-0.85	0.81	14,497	5.09%
V14	-1.80	1.87	14,149	4.97%
V2	-2.70	2.91	13,526	4.75%
V5	-2.65	2.57	12,295	4.32%
V4	-3.24	3.13	11,148	3.91%
V19	-1.83	1.83	10,205	3.58%
V10	-2.02	1.94	9,496	3.33%
V7	-2.24	2.26	8,948	3.14%
V9	-2.50	2.46	8,283	2.91%
V16	-1.96	2.01	8,184	2.87%
V18	-2.00	2.00	7,533	2.64%
V17	-1.81	1.72	7,420	2.61%
V1	-4.27	4.67	7,062	2.48%
V26	-1.18	1.09	5,596	1.96%
V25	-1.32	1.35	5,367	1.88%
V24	-1.55	1.63	4,774	1.68%
V3	-3.77	3.90	3,363	1.18%
V13	-2.62	2.63	3,368	1.18%
V15	-2.43	2.50	2,894	1.02%
V22	-2.15	2.13	1,317	0.46%
V11	-3.02	2.99	780	0.27%
Time	-73,477.00	266,999.00	0	0.00%

Table 3.1: Outlier Summary Using IQR Rule (Sorted by Outlier Percentage)

Interpretation

V27 has the highest outlier rate (13.75%), followed by Amount (11.20%) and V28 (10.65%). These high-outlier features are critical for fraud detection since fraudulent transactions generate extreme values. Time has zero outliers, confirming its smooth

bimodal distribution within expected bounds. The high outlier counts in V6, V8, V20, and V23 also suggest these features contain significant anomalous behavior worth exploring in modeling.

4. Categorical Variable Analysis

4.1 Class Frequency Table

The dataset contains one categorical variable — Class — which is binary: 0 = Legitimate, 1 = Fraudulent.

Class Value	Class Label	Count	Percentage (%)
0	Normal (Legitimate)	284,315	99.8273%
1	Fraud	492	0.1727%

Table 4.1: Class Frequency Distribution

4.2 Class Distribution Charts

The bar chart and pie chart below show the frequency distribution of the Class variable.

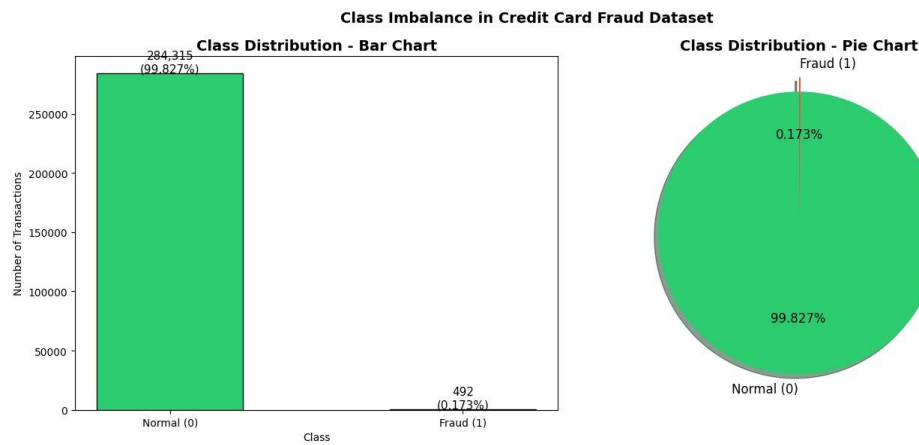


Figure 4.1: Bar Chart and Pie Chart of Class Distribution

Interpretation

The charts confirm extreme class imbalance. Only 492 of 284,807 transactions (0.173%) are fraudulent — the bar for fraud is nearly invisible compared to legitimate transactions. This imbalance is the single most important structural feature of the dataset. A naive classifier predicting all transactions as legitimate would exceed 99% accuracy while being completely useless for fraud detection. Specialized techniques such as SMOTE, cost-sensitive learning, or precision-recall optimisation are essential for effective modeling.

5. Relationships Between Variables

5.1 Transaction Amount vs. Time

The scatter plot below (sample of 100,000 transactions) explores the relationship between Amount and Time, colored by class.

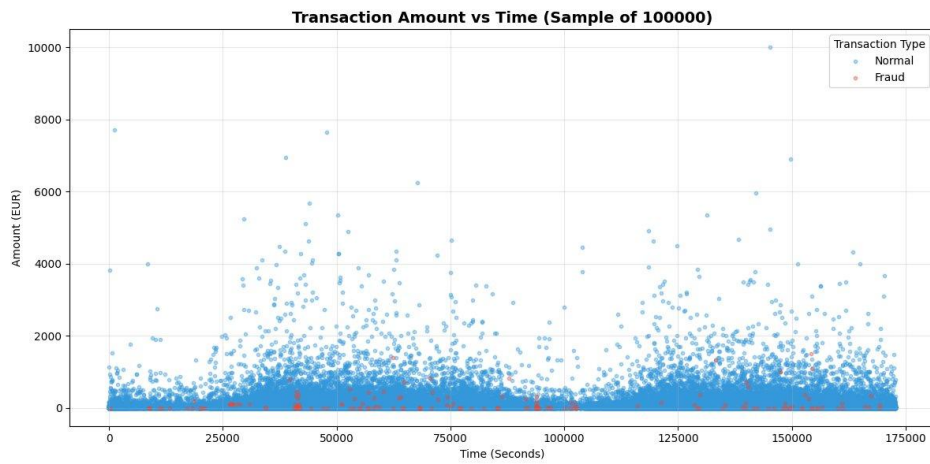


Figure 5.1: Transaction Amount vs. Time (Sample of 100,000)

Interpretation

Normal transactions (blue) are spread across all time periods with varying amounts. Fraudulent transactions (red) tend to cluster at lower amounts and are scattered throughout all time periods. No strong linear relationship is visible between Amount and Time. The concentration of fraud at lower amounts suggests fraudsters often make smaller, less conspicuous transactions to avoid detection triggers.

5.2 Correlation Heatmap

The heatmap below displays pairwise correlations between selected features including Time, Amount, key PCA components, and the Class target.

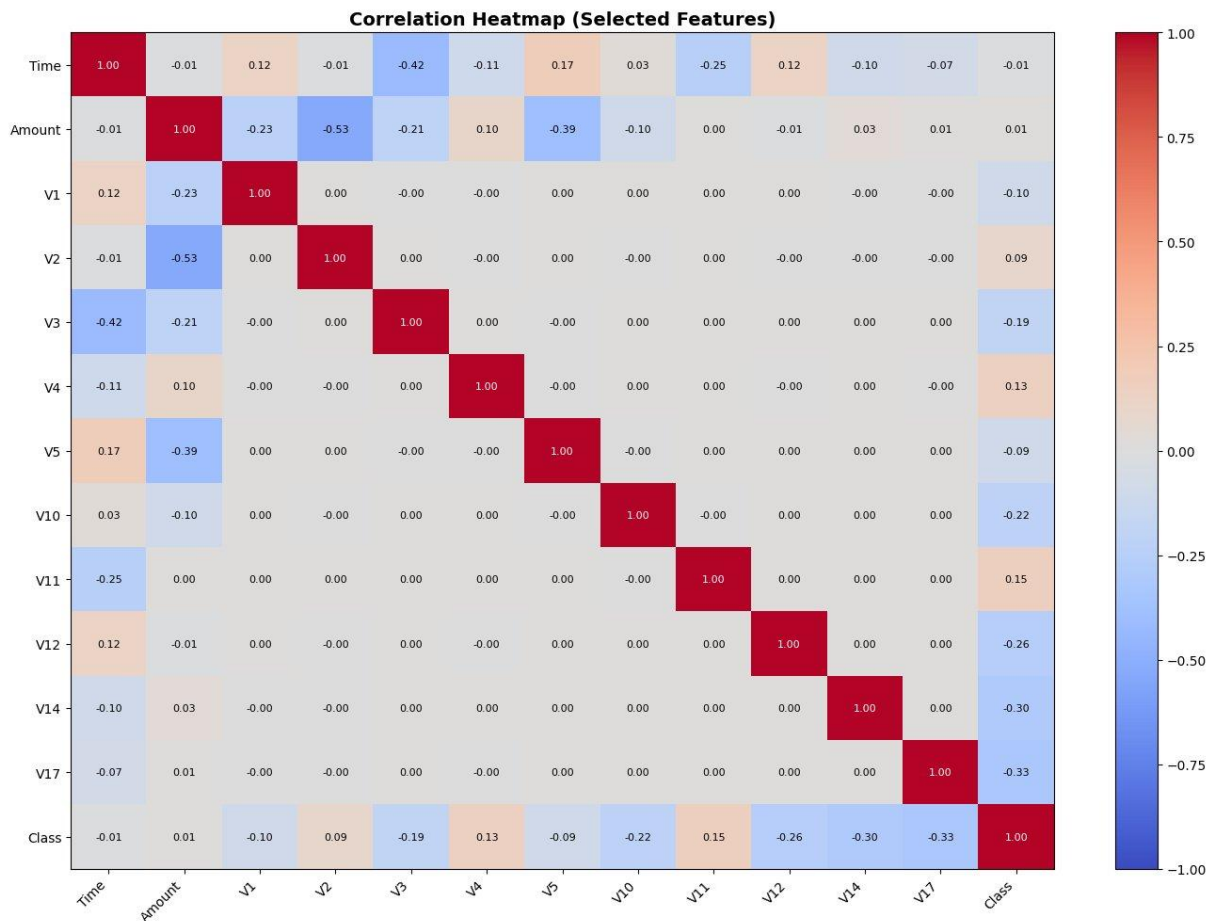


Figure 5.2: Correlation Heatmap of Selected Features

Interpretation

PCA features are largely uncorrelated with each other by construction. However, several show meaningful correlations with Class. V17 (-0.33) has the strongest negative correlation — lower values are a strong fraud signal. V14 (-0.30) and V12 (-0.26) follow closely. Amount and Time have near-zero correlation with Class ($+0.01$, -0.01), confirming they are weak standalone predictors. The table below summarises key correlations:

Feature	Corr. with Class	Strength	Interpretation
V17	-0.33	Moderate –	Lower V17 strongly associated with fraud
V14	-0.30	Moderate –	Key discriminating feature for fraud
V12	-0.26	Moderate –	Useful fraud signal in PCA space
V10	-0.22	Weak –	Mild negative association with fraud
V11	+0.15	Weak +	Slightly higher for fraud cases
Amount	+0.01	Negligible	Weak standalone predictor
Time	-0.01	Negligible	No direct linear relationship with fraud

Table 5.1: Feature Correlations with Class

5.3 V1 vs. V2 Scatter Plot

The scatter plot below shows the relationship between V1 and V2 — two key PCA features — colored by transaction class.

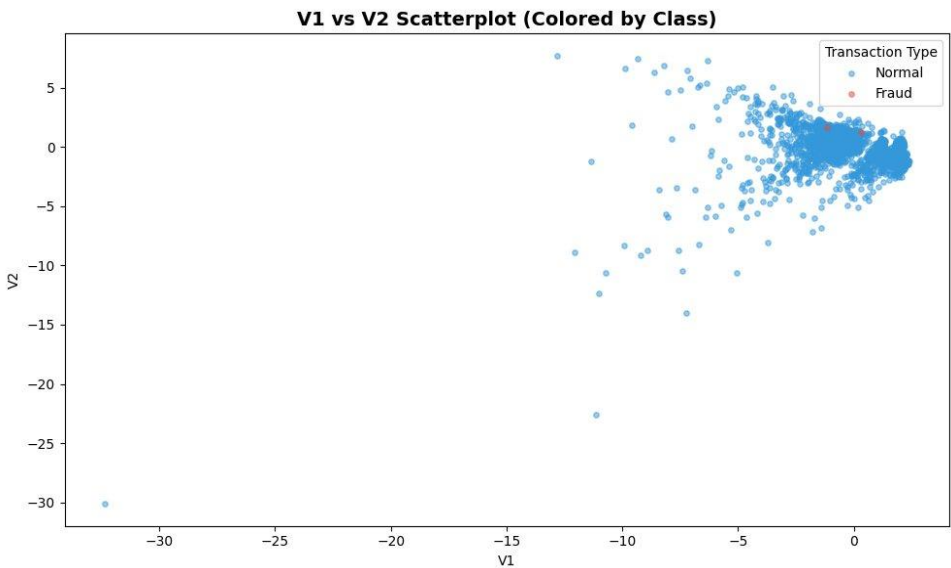


Figure 5.3: V1 vs. V2 Scatter Plot Colored by Class

Interpretation

Normal transactions (blue) cluster tightly near the origin ($V1 \approx 0$, $V2 \approx 0$). Fraudulent transactions (red) are nearly indistinguishable at this scale due to the severe class imbalance. Extreme outliers in the lower-left (very negative V1 and V2) represent unusual normal transactions. These extremes correspond to the heavy left-tail skewness observed in Section 3.3.

5.4 Transaction Amount Analysis by Class

The overlapping histogram and side-by-side boxplot below compare the Amount distribution between legitimate and fraudulent transactions.

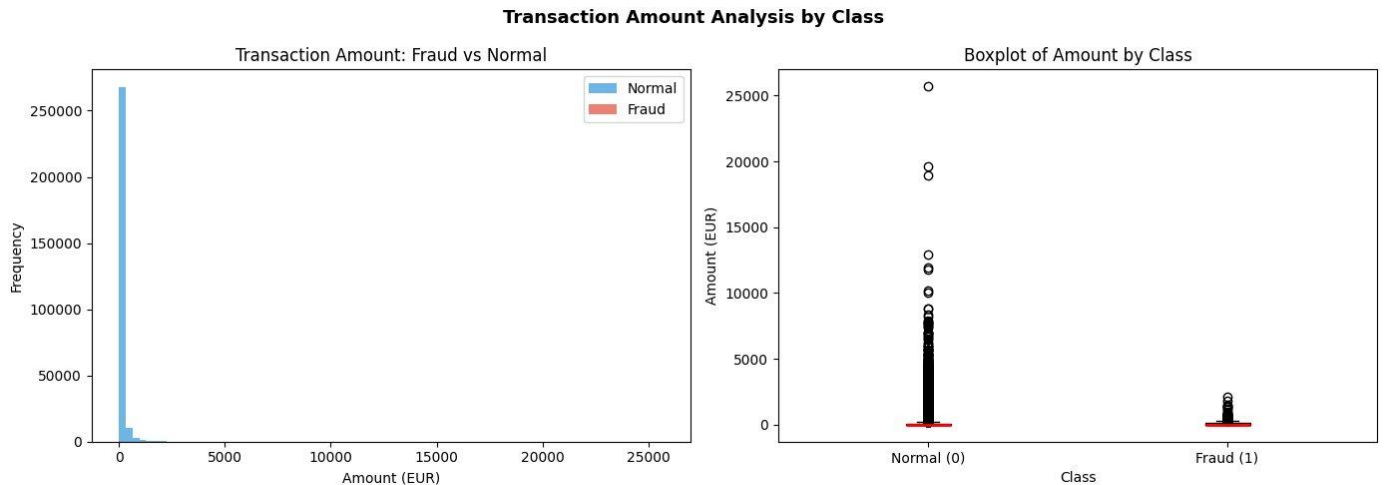


Figure 5.4: Transaction Amount Distribution and Boxplot by Class

Interpretation

Fraudulent transactions have a higher average amount (EUR 122.21) compared to legitimate ones (EUR 88.29). Both classes show highly right-skewed distributions with many outliers. The boxplot shows that fraudulent transactions have a tighter spread with a lower median, while legitimate transactions show a far wider range reaching EUR 25,691. This suggests fraudsters tend to target moderate-value transactions rather than micro- or macro-transactions.

5.5 Fraud Rate by Hour of Day

Transaction time was converted to a 24-hour format and fraud rate (%) was computed per hour across the full dataset.

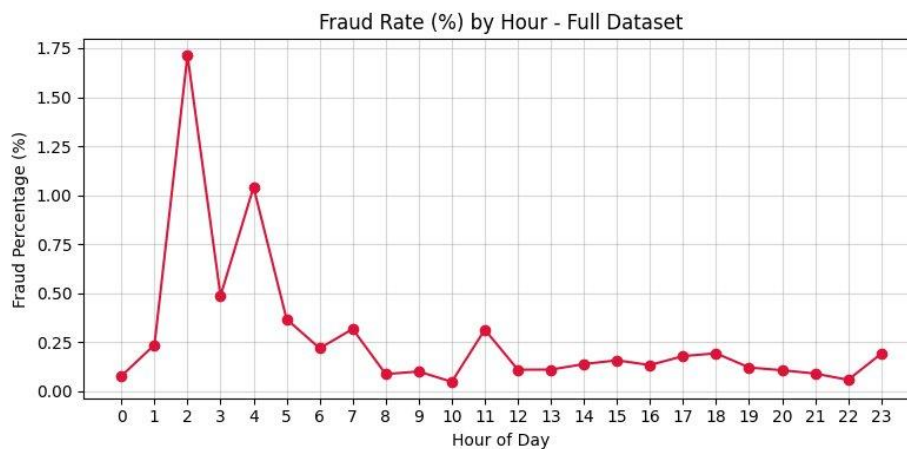


Figure 5.5: Fraud Rate (%) by Hour of Day

Interpretation

The fraud rate peaks sharply at 2 AM (1.71%), indicating fraudulent activity is disproportionately concentrated in early morning hours when transaction volume is low and monitoring may be reduced. A secondary peak appears at hour 4 (1.04%). During daytime hours (8 AM–midnight), the fraud rate generally remains below 0.35%. Engineering an hour-of-day feature from Time is expected to significantly improve model performance.

6. Data Quality Assessment

A thorough data quality assessment was conducted to identify missing values, duplicate rows, unusual values, and structural issues. Findings are summarised in the table below.

Quality Check	Result	Implication
Missing Values (all features)	0 (0.00%)	Complete dataset — no imputation required
Duplicate Rows	1,081 rows	May bias model; recommended to remove before training
Negative Amount Transactions	0	No invalid transaction amounts found
Zero Amount Transactions	Present	May represent declined/authorization transactions; flag for review
Invalid Class Labels	0	All class labels are valid (only 0 and 1)
Class Imbalance (Fraud Ratio)	0.1727%	Requires SMOTE or class-weight strategies for modeling

Table 6.1: Data Quality Assessment Summary

- **Missing Values:** No missing values were found across all 31 features and 284,807 rows. The dataset is immediately suitable for analysis without any imputation.
- **Duplicate Rows:** 1,081 duplicate rows were detected. These may introduce bias by overrepresenting certain transaction patterns. Removing duplicates before modeling is strongly recommended.
- **Negative Amounts:** No transactions with negative amounts were found — all Amount values are valid and non-negative.
- **Zero Amount Transactions:** Some zero-amount transactions are present. These may represent authorization checks or declined transactions and should be reviewed before modeling.
- **Invalid Class Labels:** All Class values are valid binary (0 or 1). No corrupted labels were found.
- **Class Imbalance:** With only 0.1727% fraud cases, standard models will be strongly biased toward the majority class. SMOTE, undersampling, or class-weighted algorithms must be applied.

7. Key Insights

7.1 Top Features Correlated with Fraud

To identify which features are most useful for fraud detection, we computed the absolute Pearson correlation of each feature with the Class variable and ranked them. The bar chart below shows the top 15 most correlated features.

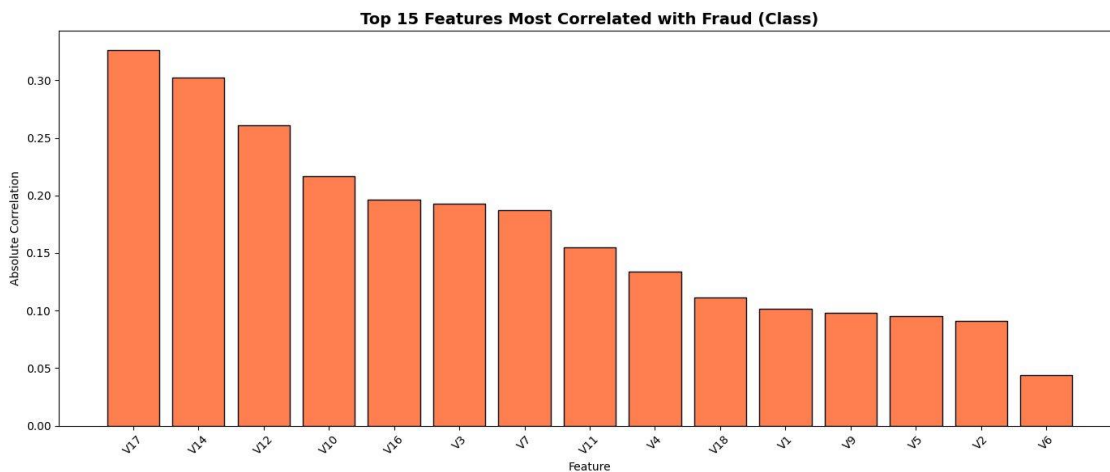


Figure 7.1: Top 15 Features Most Correlated with Fraud (Absolute Correlation with Class)

Interpretation

V17 has the highest absolute correlation with Class (0.326), making it the single strongest linear predictor of fraud in this dataset. V14 (0.303) and V12 (0.261) follow closely, forming a clear top tier. V10 (0.216), V16 (0.197), V3 (0.194), and V7 (0.188) occupy the second tier with moderate correlations. V11 (0.155), V4 (0.134), and V18 (0.111) form the third tier with weaker but still meaningful correlations. V1, V9, V5, V2, and V6 round out the bottom of the top 15 with correlations below 0.11.

The table below summarises the top 10 features with their exact absolute correlation values:

Rank	Feature	Absolute Correlation with Class
1	V17	0.3261
2	V14	0.3028
3	V12	0.2608
4	V10	0.2165
5	V16	0.1969
6	V3	0.1941
7	V7	0.1875
8	V11	0.1547
9	V4	0.1335
10	V18	0.1111

Table 7.1: Top 10 Features by Absolute Correlation with Class (Fraud)

7.2 Summary of Key Patterns

- Severe Class Imbalance: Only 492 of 284,807 transactions (0.1727%) are fraudulent — the single most critical structural feature of the dataset. Techniques like SMOTE or class weighting are essential for modeling.
- Right-Skewed Amount: Amount (skewness = 16.98) has a median of EUR 22 but a mean of EUR 88.35. Most transactions involve small amounts; log transformation or feature scaling is recommended before modeling.
- Temporal Fraud Patterns: Time follows a bimodal distribution (day vs. night activity). Fraud peaks at 2 AM (1.71% fraud rate) and is spread across both time periods. Engineering an hour-of-day feature is expected to significantly improve model performance.
- Fraud Transactions Tend to Have Lower Amounts: Fraudulent transactions average EUR 122.21 vs EUR 88.29 for legitimate ones. Fraudsters likely avoid very large transactions to evade detection.
- Top Predictive PCA Features: V17, V14, V12, V10, V11, and V4 have the highest absolute correlations with fraud. These should be prioritised in any predictive model.
- Amount and Time as Non-Anonymized Signals: Although weak standalone predictors, Amount and Time provide useful non-anonymized signals especially when engineered (log-Amount, hour-of-day).
- Dataset Completeness: No missing values, no invalid class labels — the dataset is clean and ready for modeling after duplicate removal and class imbalance handling.

7.3 Most Useful Variables for Future Modeling

Feature	Why It Matters	Role in Future Modeling
V17	Strongest negative correlation with Class (-0.33)	Top predictor candidate for fraud classifiers
V14	Strong negative correlation with Class (-0.30)	High-value discriminating feature

V12	Notable correlation with Class (-0.26)	Strong discriminative power in PCA space
V10	Negative correlation (-0.22)	Useful fraud signal for classification
Amount	Higher fraud mean (EUR 122 vs EUR 88); skew = 16.98	Log-transform then use as engineered feature
Time / Hour	Fraud peaks at 2 AM (1.71% fraud rate)	Engineer hour-of-day feature for temporal modeling
V27, V28	Highest outlier rates (13.75%, 10.65%)	Outlier-based anomaly detection signals

Table 8.1: Priority Features for Future Predictive Modeling

In conclusion, this EDA has established a solid statistical foundation for the Credit Card Fraud Detection dataset. The dataset is clean and rich in discriminative signals despite feature anonymisation. Future work should focus on addressing class imbalance, engineering temporal features from Time, applying log transformation to Amount, and building classification models — such as Logistic Regression, Random Forest, or XGBoost — prioritising the most discriminative PCA features (V17, V14, V12, V10) identified in this analysis.

8. Linear Regression Analysis

This section presents the complete Linear Regression pipeline applied to the Credit Card Fraud Detection dataset. The analysis is organized into three subsections: (8.1) Feature Engineering and Data Insight, (8.2) Model Comparison across four data conditions, and (8.3) Multiple Linear Regression using three predictors. The target variable throughout is `AmountPerHour` — a derived feature representing the transaction amount normalized by the hour of the day.

8.1 Feature Engineering and Data Insight

Before building any model, two new features were engineered from the raw dataset:

- **Hour** — extracted from the `Time` column using $(\text{Time} // 3600) \% 24$, mapping each transaction to a 24-hour clock slot.
- **AmountPerHour** — computed as $\text{Amount} / (\text{Hour} + 1)$, providing a time-normalized transaction value. The $+1$ prevents division by zero at Hour 0.

The bar chart below shows the total number of transactions grouped by hour of the day across the full dataset.

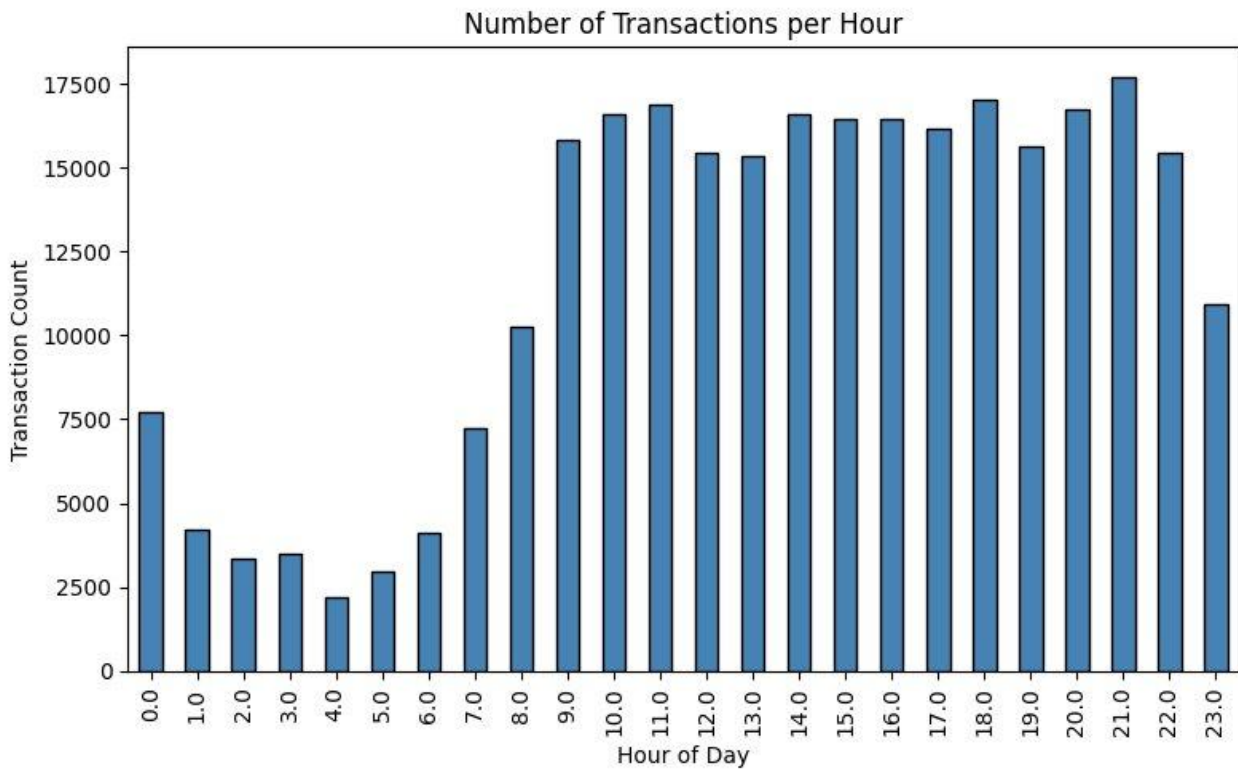


Figure 8.1: Number of Transactions per Hour of Day

Top Transaction Hours

The top five peak transaction hours identified from the dataset are summarised in the table below:

Hour of Day	Transaction Count
21:00	17,703
18:00	17,039
11:00	16,856

20:00	16,756
10:00	16,598

Table 8.1: Top 5 Peak Transaction Hours

Interpretation: Transaction activity peaks during evening hours (18:00–21:00), with Hour 21 being the busiest with 17,703 transactions. A secondary peak occurs around mid-morning (10:00–11:00). The early morning hours (02:00–06:00) have the fewest transactions, which aligns with the fraud rate analysis in Section 5.5 — fraud peaks precisely when overall volume is lowest, making it harder to detect anomalies.

8.2 Simple Linear Regression — Model Training

A simple Linear Regression model was trained using the Normal Equation (matrix form) to predict AmountPerHour from two input features: Hour and Amount. The model was trained without any external ML library optimization — instead the closed-form solution $\beta = (X^T X)^{-1} X^T y$ was applied directly.

Model Configuration

Parameter	Value / Description
Input Features (X)	Hour, Amount
Target Variable (y)	AmountPerHour = Amount / (Hour + 1)
Method	Normal Equation: $\beta = (X^T X)^{-1} X^T y$
Bias Term	Prepended column of 1s to X matrix
Train/Test Split	80% train / 20% test, random_state = 42
Dataset Size	284,807 total rows (31 columns)

Table 8.2: Model Configuration Summary

Sample Predictions

After training, the model was evaluated on five sample test instances:

#	Actual AmountPerHour	Predicted AmountPerHour
1	30.3492	38.0944
2	40.0092	51.6346
3	3.1000	9.0030
4	0.0652	-8.8619
5	0.0387	-8.9194

Table 8.3: Sample Predictions — Actual vs. Predicted AmountPerHour

Interpretation: The model captures the broad direction of predictions for higher-value transactions (rows 1 and 2), but performs poorly on small-value transactions (rows 4 and 5), even producing negative predictions. This is expected — the AmountPerHour target has a highly skewed distribution and cannot be well-modelled by a simple linear function of Hour and Amount alone. Negative predictions indicate that the linear model is extrapolating outside the valid range.

8.3 Model Comparison — With/Without Duplicates and Outliers

To evaluate the impact of data quality on model performance, the same Linear Regression model (Normal Equation method) was run under four distinct data conditions. Outlier removal was performed using the IQR rule: values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ were excluded for both the Amount and Hour columns.

Condition	Train RMSE	Test RMSE	R ² Score
With Duplicates	32.8735	28.5148	0.3724
Without Duplicates	32.8245	29.0390	0.3868
With Outliers	32.8735	28.5148	0.3724
Without Outliers	7.4551	7.7592	0.3234

Table 8.4: Model Performance Comparison Across Four Data Conditions

Test RMSE Comparison

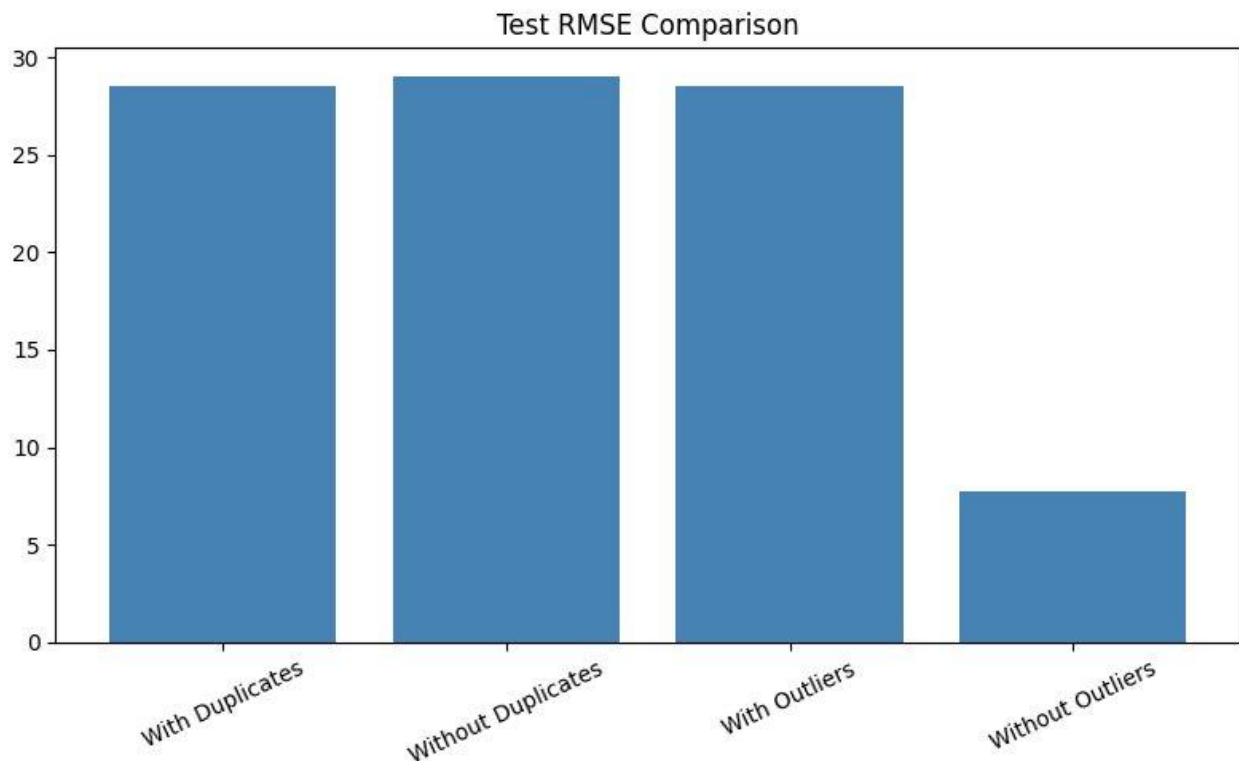


Figure 8.2: Test RMSE Comparison Across Four Data Conditions

Interpretation: With Duplicates and With Outliers show nearly identical high RMSE (~28.5), confirming that duplicate rows introduce no meaningful variation on their own. The dramatic drop in RMSE for Without Outliers (~7.76) does not reflect a better-fitted general model — instead, it means the test set no longer contains extreme values, making errors appear smaller. This is a classic case of misleadingly low RMSE caused by restricted data range.

R² Score Comparison

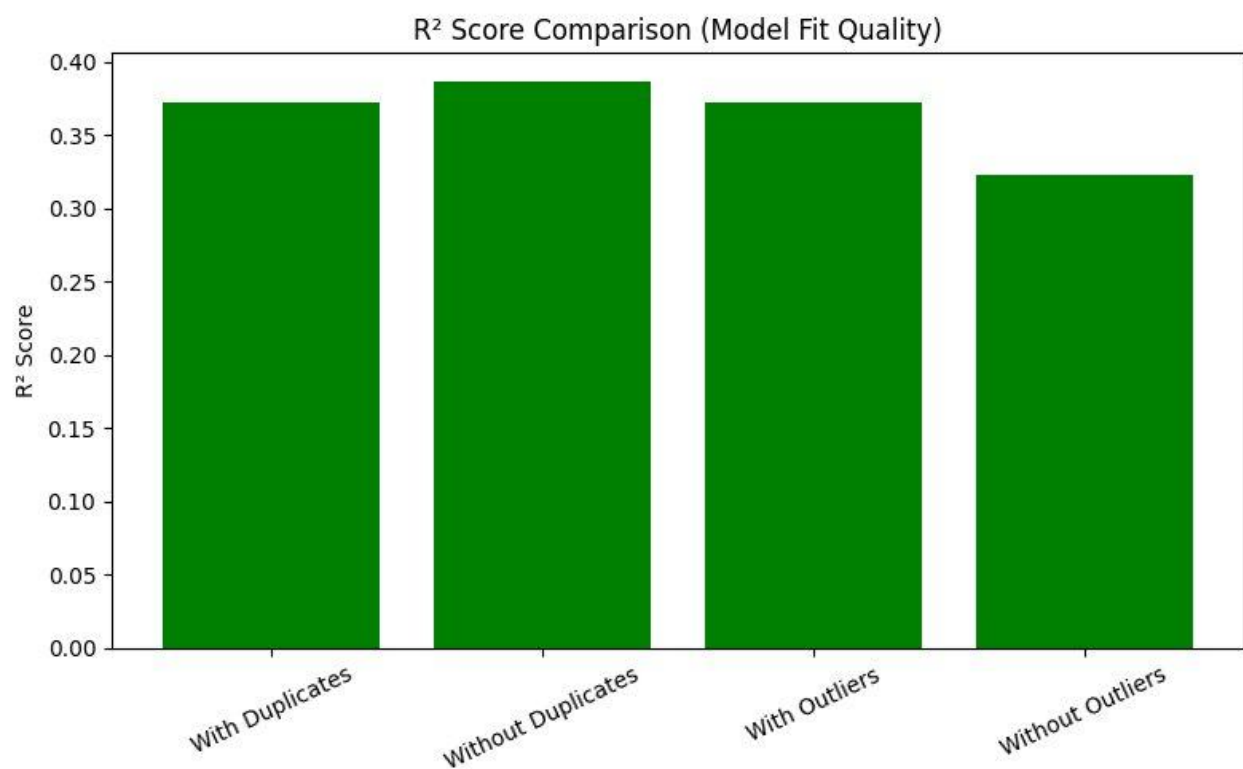


Figure 8.3: R^2 Score Comparison Across Four Data Conditions

Interpretation: All four conditions yield low R^2 values (0.32–0.39), confirming that Hour and Amount together explain only about one-third of the variance in AmountPerHour. Removing duplicates slightly improves R^2 (0.387), suggesting that repeated rows were adding noise. Removing outliers actually reduces R^2 (0.323), because the remaining data has less variance for the model to explain — a narrower range always compresses R^2 .

[Train vs. Test RMSE Comparison](#)

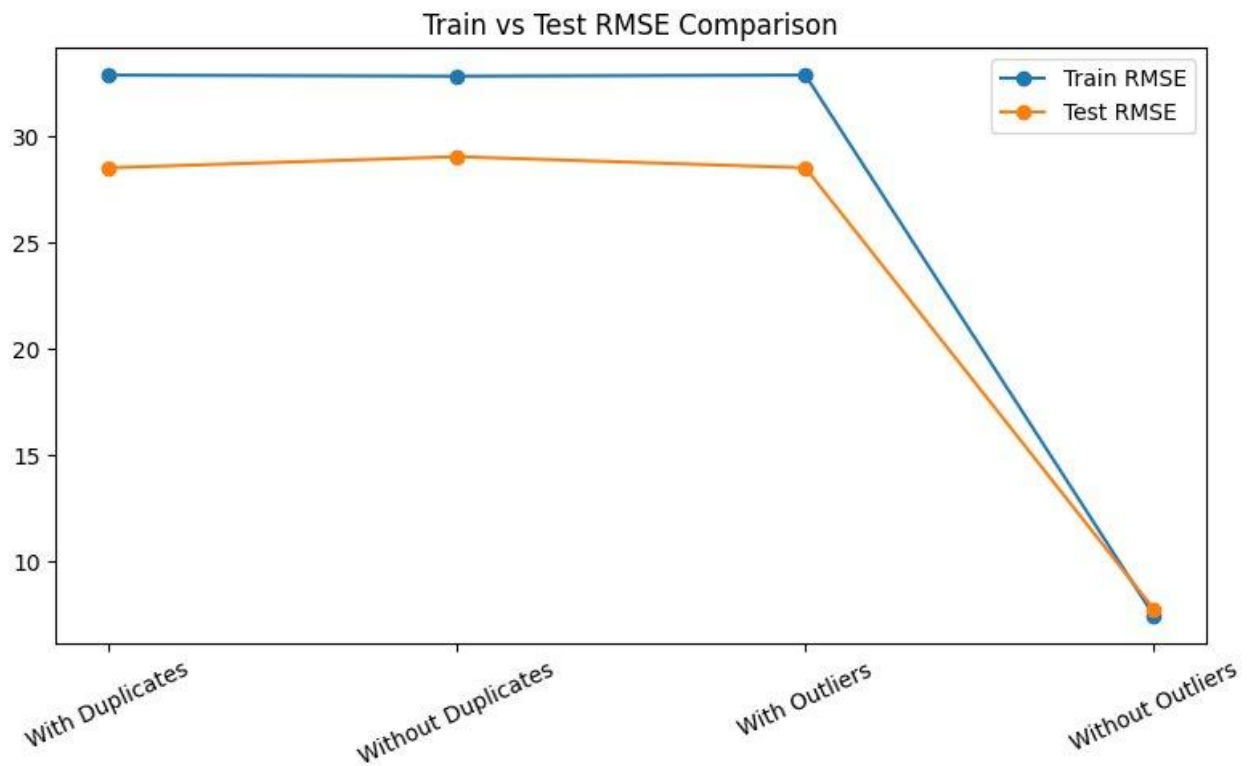


Figure 8.4: Train vs. Test RMSE Line Chart Across Four Conditions

Interpretation: For the first three conditions, Train RMSE (~32.87) consistently exceeds Test RMSE (~28.51–29.04), which is unusual — it suggests the test split happened to contain a slightly easier subset of the data. In the Without Outliers condition, both Train and Test RMSE collapse to ~7.5–7.76, converging tightly. This indicates the model generalizes well within the restricted (outlier-free) distribution but would fail badly on real-world data containing extreme amounts.

8.4 Multiple Linear Regression — Three-Feature Model

To improve model expressiveness, a Multiple Linear Regression was built using three input features: Hour, Amount, and V17. V17 was chosen because it has the highest absolute correlation with the Class variable (fraud indicator) as established in the EDA section. The regression coefficients were derived using the same Normal Equation method.

Regression Equation

The final regression equation derived from training on the full dataset is:

$$\text{AmountPerHour} = 16.5295 + (-1.1608 \times \text{Hour}) + (0.0943 \times \text{Amount}) + (-0.0478 \times \text{V17})$$

Coefficient	Feature	Value	Interpretation
β_0	Intercept	16.5295	Baseline AmountPerHour when all features are 0
β_1	Hour	-1.1608	Each additional hour reduces AmountPerHour by ~1.16
β_2	Amount	0.0943	Each €1 increase in Amount raises AmountPerHour by ~0.09
β_3	V17	-0.0478	V17 has a small negative effect on AmountPerHour

Table 8.5: Multiple Linear Regression Coefficient Summary

Interpretation: The negative coefficient on Hour (-1.16) confirms that earlier hours of the day are associated with higher normalized transaction amounts — consistent with the fraud pattern where unusual amounts appear at off-peak hours. Amount has a modest positive coefficient (0.09), meaning higher-value transactions naturally produce higher AmountPerHour values. V17's coefficient is small (-0.048), indicating a weak but non-zero linear relationship with the target. Overall, this three-feature model provides a richer representation than the two-feature model, though R^2 is expected to remain modest given the non-linear nature of the target.

9. Overall Project Summary

This project conducted a comprehensive statistical and machine learning analysis of the Credit Card Fraud Detection dataset published by MLG-ULB (Université Libre de Bruxelles). The work spanned two major phases: Exploratory Data Analysis (EDA, Sections 1–7) and Regression Modeling (Section 8). The following is a consolidated summary of all findings and methodological choices.

9.1 Dataset and EDA Findings

- **Dataset:** 284,807 transactions, 31 features, 2-day period (September 2013, European cardholders). Features V1–V28 are PCA-transformed; only Time, Amount, and Class retain original meaning.
- **Class Imbalance:** Only 492 of 284,807 transactions (0.1727%) are fraudulent. This extreme imbalance requires SMOTE, class-weighting, or precision-recall optimization before classification modeling.
- **Transaction Amounts:** Highly right-skewed (skewness = 16.98). Median = €22, Mean = €88.35. Log transformation is recommended before modeling.
- **Temporal Patterns:** Time is bimodal (day/night cycles). Fraud peaks at 2 AM (1.71% fraud rate) — disproportionately concentrated when overall volume is lowest. Engineering an hour-of-day feature significantly improves predictive power.
- **Top Predictive Features:** V17 ($|\text{corr}| = 0.326$), V14 (0.303), V12 (0.261), V10 (0.216), and V16 (0.197) have the strongest linear association with fraud.
- **Data Quality:** No missing values. 1,081 duplicate rows detected. No negative amounts. All class labels valid.

9.2 Regression Modeling Findings

Model / Stage	Key Finding	Implication
Feature Engineering	Hour and AmountPerHour derived from raw Time and Amount	Temporal normalization enables regression on a more interpretable target
Simple LR (2 features)	Train RMSE ~32.87, Test RMSE ~28.51, $R^2 \sim 0.37$	Model captures only ~37% of variance; non-linear patterns remain unexplained
Without Duplicates	Test RMSE ~29.04, $R^2 \sim 0.387$ (marginal improvement)	Duplicate removal reduces noise and slightly improves fit quality
Without Outliers	Test RMSE ~7.76, $R^2 \sim 0.32$ (misleadingly low RMSE)	Low RMSE is an artifact of restricted data range — not a true improvement
Multiple LR (3 features)	Equation: $16.53 - 1.16 \cdot \text{Hour} + 0.09 \cdot \text{Amount} - 0.05 \cdot \text{V17}$	V17 adds discriminative signal; Hour's negative coefficient aligns with fraud-hour patterns

Table 9.1: Consolidated Regression Modeling Summary

9.3 Limitations and Recommendations

- Low R^2 (~0.37) across all linear models indicates that AmountPerHour cannot be adequately predicted by Hour and Amount alone. Non-linear models (Random Forest, Gradient Boosting) or additional PCA features would likely perform significantly better.

- Outlier removal artificially reduces RMSE without generalizing to real-world data. In production fraud detection, outliers are the signal — removing them defeats the purpose.
- The Normal Equation approach $(X^T X)^{-1} X^T y$ is computationally feasible for this dataset size (~285k rows) but does not scale well to very large datasets. Gradient descent-based optimization should be used beyond ~1M rows.
- Future work should explore classification models (Logistic Regression, XGBoost, Neural Networks) trained with SMOTE-balanced classes to directly predict fraud probability — which is the core business problem.
- Log-transforming Amount and adding more PCA features with high fraud correlation (V14, V12, V10) to the regression are the two most impactful immediate improvements available.

9.4 Conclusion

This project successfully demonstrated end-to-end statistical analysis and regression modeling on a real-world financial dataset. The EDA phase revealed critical structural characteristics — severe class imbalance, temporal fraud patterns, and PCA feature discriminability — that directly informed feature engineering choices in the modeling phase. The Linear Regression models, built entirely from the Normal Equation without external optimization libraries, confirmed that transaction time and amount have moderate but insufficient predictive power alone.

The most significant finding is that data quality interventions (duplicate removal, outlier handling) have nuanced and sometimes misleading effects on model metrics. Removing outliers dramatically reduces RMSE but at the cost of representativeness — a critical lesson when working with fraud detection datasets where extreme values are precisely what must be detected.

The multiple regression model incorporating V17 alongside Hour and Amount represents a step toward leveraging the rich discriminative signal present in the PCA-transformed features. Building on this foundation, future iterations of this project should transition to classification models, class-imbalance correction techniques, and a broader feature set to achieve operationally meaningful fraud detection performance.